3.4 Compositions of Transformations

A composition applies one rigid motion, then another, to build a single combined transformation — and the order in which you apply them can change the final image, which surprises many students used to addition being commutative. This section proves two elegant theorems (two reflections over parallel lines make a translation; two reflections over intersecting lines make a rotation through double the angle between them), defines the glide reflection precisely, and works both forward (apply a sequence) and backward (find the missing vector or angle from a known result).

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Does composition order always commute?

2. $(1, 2)$ by $\langle 0, 3\rangle$, then over y-axis.

3. Translation + reflection over a parallel line is a what?

4. Equivalent to a what?

5. Equivalent to a what?

6. Two successive $90°$ CCW about origin = ?

7. Composition of isometries is an isometry?

8. $(3, 1)$: $180°$ about origin, then reflect over x-axis.

9. $(2, 4)$: reflect over x-axis, then over y-axis.

10. Reflect over both axes (either order) equals what rotation?

11. $\langle 2, 0\rangle$ then $\langle 0, 5\rangle$ equals what vector?

12. Glide reflection reverses orientation?

13. Reflect twice over the same line. Result?

14. Reflect $(5, -2)$ over $y = x$, then reflect that image over the x-axis. Find the result and identify the equivalent rotation.

15. A composition of "rotate $90°$ CCW about the origin, then translate" maps $(1, 0)$ to $(4, 6)$. Find the translation vector, then find the image of $(0, 2)$ under the same composition.

✓ Answer Key

1. No
2. $(-1, 5)$
3. Glide reflection
4. Translation
5. Rotation
6. $180°$ rotation
7. Yes
8. $(-3, 1)$
9. $(-2, -4)$
10. $180°$
11. $\langle 2, 5 \rangle$
12. Yes
13. Identity (original figure)
14. $(-2, -5)$; a $90°$ clockwise rotation about the origin (twice the $45°$ angle between the lines)
15. vector $\langle 4, 5 \rangle$; image of $(0, 2)$ is $(2, 5)$